Budget-Aware Controller Blending: Post-Hoc Damage-Budget Enforcement for Pre-Trained RL Policies

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Abstract

Deploying RL policies under cumulative *damage budgets*—limits on fatigue load, hazard exposure, or any integrated operational cost—usually requires retraining for each budget specification. We propose a closed-form schedule that *blends two pre-existing controllers at inference*: an aggressive performance policy π_{perf} and a calibrated safety controller π_{safe}. The blend weight $\sigma(u)$ is a function of the *urgency ratio* $u = \rho/\tau$ (remaining budget fraction over remaining time fraction). When $u > 1$ (on pace), $\sigma = 0$ and the performance policy runs; when $u \to 0$ (overspent), $\sigma \to 1$ and the safety controller takes over. A single scalar controls sharpness; a single policy serves arbitrary budget levels instantly. On two domains we show parallel results under the same schedule: wind-farm yaw control, where π_{perf} is an RL wake-steering policy and π_{safe} is a zero-yaw fallback—retaining 95.9±3.1% of unconstrained power across 4 seeds while satisfying a 15-step negative-yaw budget; and Safety Gymnasium, where π_{perf} is a goal-seeking SAC actor and π_{safe} is an artificial-potential-field navigator—reducing hazard entries while preserving reward. A gradient-based variant of the same schedule (correction via a learned cost critic Q_c) works when the *action-to-cost coupling strength* $\kappa = \mathbb{E}\|\nabla_a Q_c\|/\mathbb{E}\|\nabla_s Q_c\|$ is large; when κ is small, the explicit-π_{safe} blend is indicated. Both cases use the same urgency schedule.

1 Introduction

A wind-farm operator has an aggressive RL controller that maximizes power via coordinated wake steering; it was trained once, unconstrained. Six months after deployment, a bearing inspection tightens the constraint: turbine 2's yaw may be negative for at most 15 hours per year. Retraining is expensive. The operator *already has* a certified fallback controller: the OEM yaw controller, or a load-minimizing setpoint table [Siemens Gamesa, 2023]. The missing piece is a way to *pace between* the aggressive RL policy and the certified safe policy so that the cumulative constraint holds over the year, without retraining either.

Cumulative operational constraints are ubiquitous. Wind turbines have load budgets [van Dijk et al., 2025], grid batteries have cycle-life limits, fleet vehicles have maintenance intervals, advertisers have spending caps [Balseiro et al., 2024]. Unlike instantaneous safety constraints, these are *lifetime budgets*: the agent must decide *when* to spend its limited capacity in the risky regime. A fixed performance-safety trade-off parameter α in a scalarized objective $\alpha P - (1-\alpha)D$ cannot express this: α has no memory and does not self-pace.

Prior work. CMDP methods [Altman, 1999, Achiam et al., 2017, Stooke et al., 2020, Sootla et al., 2022] retrain per budget. Post-hoc composition [Urain et al., 2023, Zhang et al., 2025, Xiao et al., 2025] handles instantaneous constraints. The missing primitive is a *post-hoc, budget-aware* composition over an episode.

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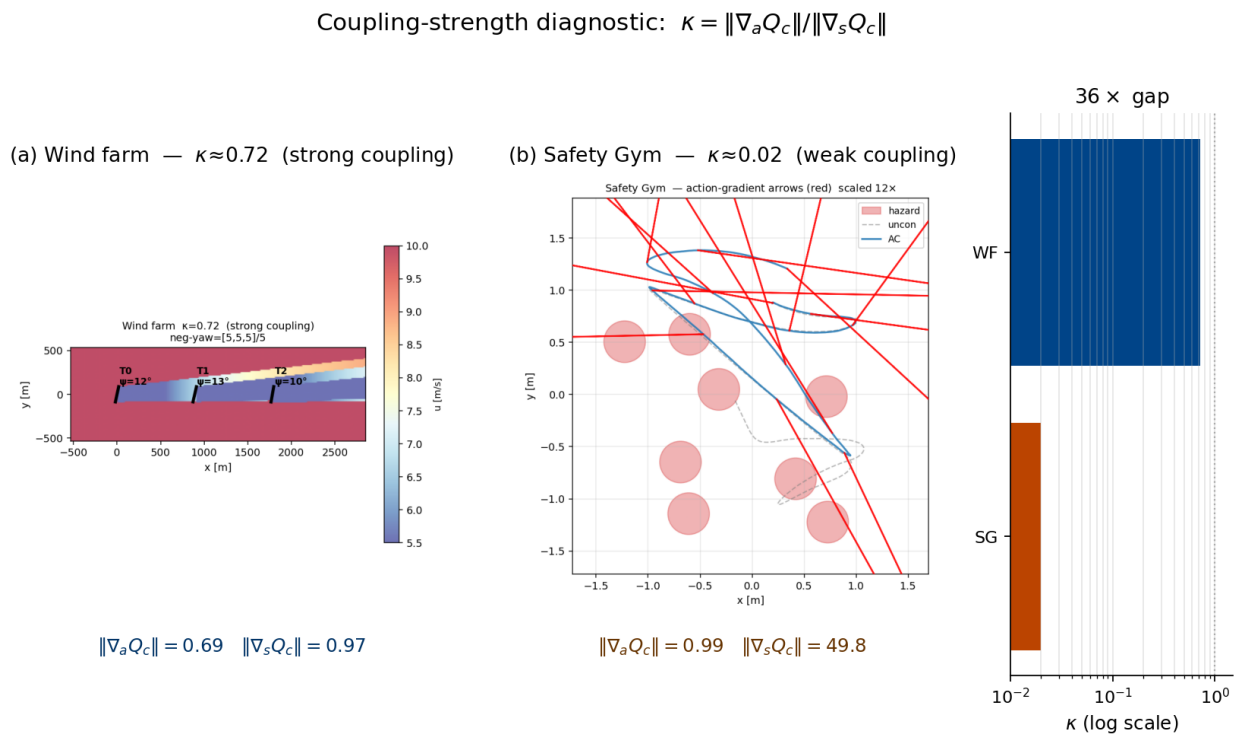

Figure 1: Two domains, two safety controllers, one urgency schedule. (a) Wind farm: π_{safe} is a zero-yaw (no wake-steering) setpoint; the action gradient $\nabla_a Q_c$ is informative ($\kappa \approx 0.72$), so a lightweight gradient-correction variant of the same schedule also works. (b) Safety Gymnasium: π_{safe} is an artificial-potential-field (APF) navigator; the action gradient is small ($\kappa \approx 0.02$), so explicit safe-controller blending is required.

Our proposal: urgency-ratio blending. We inherit the trader's intuition from optimal execution [Almgren and Chriss, 2001]: pace consumption against remaining inventory vs. remaining time. Define

$$u_t = \frac{\rho_t}{\tau_t}, \qquad \rho_t = \frac{d - C_t}{d}, \qquad \tau_t = \frac{T - t}{T}, \tag{1}$$

and blend two pre-existing controllers at inference:

$$a_t = (1 - \sigma(u_t))\, \pi_{\text{perf}}(s_t) + \sigma(u_t)\, \pi_{\text{safe}}(s_t), \tag{2}$$

where $\sigma : \mathbb{R}_{\geq 0} \to [0, 1]$ is monotone decreasing in u (Section 3): on pace $\Rightarrow$ performance; deficit $\Rightarrow$ safety.

Contributions. (1) A *closed-form urgency-ratio schedule* $\sigma(u)$ with a single sharpness parameter, derived from budget-balanced allocation under an idealized Boltzmann response (§3; Prop. 1). (2) A *two-controller composition* that treats existing safety controllers as first-class primitives rather than retraining the policy (§4). (3) A *gradient-correction variant* (§4.2) where π_{safe} is implicitly defined by a step of $-\nabla_a Q_c$; we show this variant works precisely when the coupling strength $\kappa = \mathbb{E}\|\nabla_a Q_c\| / \mathbb{E}\|\nabla_s Q_c\|$ is large, and degenerates when κ is small. (4) *Empirical validation* on wind-farm yaw control ($\kappa \approx 0.72$) and Safety Gymnasium ($\kappa \approx 0.02$), under the *same* schedule, with *different* safety controllers chosen by κ (§5).

Deployment claim. The method drops into farms that already have an RL wake-steering policy and an OEM or WakeAdapt-style safe controller; no retraining, no new certification. The safety layer is already certified; the urgency schedule decides when to fall back to it.

2 Background

Constrained MDPs. A CMDP [Altman, 1999] adds cost constraints to the MDP objective: $\max_\pi \mathbb{E}_\pi[\sum_t r_t]$ subject to $\mathbb{E}_\pi[\sum_t c_t^i] \le d^i$ per agent i [Ray et al., 2019, Achiam et al., 2017]. Lagrangian methods [Stooke et al., 2020] learn dual variables at training time; changing d requires a new policy. SAUTE [Sootla et al., 2022] augments the state with ρ_t but still retrains per budget.

Post-hoc composition. Energy composition [Urain et al., 2023, Gladstone et al., 2025], classifier guidance [Zhang et al., 2025], and CBFs [Xiao et al., 2025, Durkin et al., 2025] modify pre-trained policies at inference to satisfy *instantaneous* constraints. We extend this tradition to *cumulative* budgets via a time-varying blend weight.

Multi-objective scalarization. A fixed trade-off $\alpha P - (1-\alpha)D$ is memoryless and cannot self-pace. Our urgency schedule turns α into a time-varying $\alpha(u_t) = (1-\sigma(u_t))\,\alpha_{\text{perf}} + \sigma(u_t)\,\alpha_{\text{safe}}$, with σ determined by the closed-form schedule below.

3 Urgency-Ratio Schedule

Proposition 1 (Idealized-Boltzmann motivation)**.** *If the performance policy responds to a scalar penalty λ via $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$, then the penalty maintaining budget balance takes the multiplicative form*

$$w^*(u) = 1/u. \tag{3}$$

Proof sketch. Budget balance $(T-t)\,P(\text{risky} \mid \lambda) = d - C_t$ with the Boltzmann substitution gives $\lambda = \frac{1}{\alpha}\ln\frac{(T-t)P_0}{d-C_t}$. The multiplicative weight relative to TWAP ($u=1$) reduces to $\tau/\rho = 1/u$. Full derivation in Appendix A.1. □

Practical schedule. We use a parametric family

$$w(u) = \exp\big(\eta \cdot (\tfrac{1}{u} - 1)\big), \qquad \eta \ge 0, \tag{4}$$

which agrees with $u^{-\eta}$ within 5% for $u \in [0.7, 1.3]$ (Appendix Fig. 3) and avoids the $1/u \to \infty$ singularity at $u = 0$ via a $\lambda_{\max}$ clamp.

Blend weight. For two-controller composition (Eq. 2), we map $w(u)$ to a bounded $\sigma \in [0,1]$:

$$\sigma(u) = \begin{cases} 0 & u \ge 1, \\ 1 - \exp(-\eta(1/u - 1)) & u < 1. \end{cases} \tag{5}$$

On pace or ahead ($u \ge 1$): pure performance policy. Deficit ($u < 1$): sigmoid-like ramp toward pure safety controller, with sharpness η.

Hard-wall backstop. A steep barrier activates when budget drops below 5%: $\sigma \to 1$ for $u < 0.05$, independent of η.

4 Method

The method has three components that compose independently: **(i)** the *schedule* $\sigma(u)$ from §3 (how much safety), **(ii)** the *performance policy* π_{perf} (what we deploy when on pace), and **(iii)** the *safety controller* π_{safe} (what we revert to on deficit). **Any** π_{safe} composes with any π_{perf} under the same schedule.

4.1 Two-Controller Blend (primary formulation)

$$a_t = (1-\sigma(u_t))\,\pi_{\text{perf}}(s_t) + \sigma(u_t)\,\pi_{\text{safe}}(s_t). \tag{6}$$

Wind farm instantiation. π_{perf} = RL wake-steering policy (EBT actor, SAC training, 100k steps unconstrained). π_{safe} = zero-yaw setpoint (no wake steering). Under our per-turbine fatigue surrogate $c = (|y|/\bar{y})^p (v/v_{\text{ref}})^q$ ($p=2$, $q=3$), the cost-minimizing setpoint is $y = 0$; in deployment π_{safe} would be a WakeAdapt-style [Siemens Gamesa, 2023] lookup table that minimizes farm-level damage equivalent load (DEL).

Safety Gymnasium instantiation. π_{perf} = SAC/MLP actor (unconstrained, goal-seeking). π_{safe} = artificial-potential-field (APF) navigator: $F_{\text{goal}} = \text{norm}(g - x)$ and $F_{\text{rep}} = \sum_i k_r(x - h_i)/\|x - h_i\|^3$, combined and mapped to body-frame (thrust, turn). No learning; uses hazard positions (a privileged but fixed-per-episode quantity, exactly what CBF literature assumes [Xiao et al., 2025]).

4.2 Gradient-Correction Variant

When an *action-dependent* cost $c(s, a)$ is available (as in wind-farm fatigue), a lightweight alternative to the explicit π_{safe} is gradient correction:

$$a_t \leftarrow \pi_{\text{perf}}(s_t) - \lambda_{\text{gs}} \cdot w(u_t) \cdot \alpha_{\text{g}} \cdot \nabla_a c(s_t, a_t), \tag{7}$$

iterated for K steps. This is Eq. 6 with π_{safe} implicitly defined by a single gradient step: effectively $\pi_{\text{safe}} \approx \pi_{\text{perf}} - \alpha_{\text{g}} \nabla_a c$. When c is learned (Q_c), this becomes the energy-composition form of Urain et al. [2023], with our urgency weight in place of a fixed coefficient.

4.3 When Does Each Variant Work?

The gradient variant (Eq. 7) requires a non-trivial $\nabla_a c$ or $\nabla_a Q_c$. We quantify this via the *coupling strength*

$$\kappa := \mathbb{E}_{(s,a)\sim\pi} \|\nabla_a Q_c(s, a)\|_2 / \mathbb{E}_{(s,a)\sim\pi} \|\nabla_s Q_c(s, a)\|_2. \tag{8}$$

κ large: action has strong leverage over future cost, gradient correction of π_{perf} suffices. κ small: cost is state-dominated, $\nabla_a Q_c$ is weak, and *explicit* π_{safe} is required.

We measure $\kappa_{\text{windfarm}} = 0.72$ and $\kappa_{\text{SG}} = 0.02$ from 2000 on-policy samples per domain (Table 1). The 36× gap predicts *which* variant to use, not *whether* the framework applies.

Table 1: Coupling strength κ per domain (2000 on-policy samples). Large $\kappa \Rightarrow$ gradient correction suffices (wind farm). Small $\kappa \Rightarrow$ explicit π_{safe} blend required (Safety Gym).

Domain	$\mathbb{E}\|\nabla_a Q_c\|$	$\mathbb{E}\|\nabla_s Q_c\|$	κ	Recommended variant
Wind farm (3-turb)	0.69±0.34	0.97±0.26	**0.72**	gradient correction (Eq. 7)
Safety Gym (PointGoal)	0.99±0.69	49.8±12.5	**0.02**	explicit π_{safe} blend (Eq. 6)

4.4 Algorithm

Cost: one extra controller evaluation per step; schedule is closed-form (negligible). Calibration: $\eta \in [1, 10]$ tuned on a handful of evaluation episodes. Unlike a Lagrangian dual μ learned via gradient ascent [Stooke et al., 2020], $\sigma(u)$ adapts *within* an episode to the realized damage trajectory.

5 Experiments

We test the same schedule on two domains with different safety controllers.

Algorithm 1 Urgency-Ratio Budget-Aware Controller Blending

Require: Performance policy π_{perf}, safety controller π_{safe}, limit d, horizon T, sharpness η

$C_0 \leftarrow 0$
for $t = 0, 1, \ldots, T-1$ **do**
  $u_t \leftarrow \frac{(d-C_t)/d}{(T-t)/T}$
  $\sigma_t \leftarrow 0$ if $u_t \geq 1$ else $1 - \exp(-\eta(1/u_t - 1))$
  $a_t \leftarrow (1-\sigma_t)\,\pi_{\text{perf}}(s_t) + \sigma_t\,\pi_{\text{safe}}(s_t)$
  Execute a_t; observe c_t; $C_{t+1} \leftarrow C_t + c_t$
end for

5.1 Wind Farm: Gradient Correction and Zero-Yaw Blend

Environment. WindGym [Nilsen et al., 2025] with PyWake [Pedersen et al., 2023] backend; 3-turbine row; wind speed 10–14 m/s, direction 225–315°. Budget d: total timesteps per turbine at negative yaw per episode (200-step episode; cf. per-year budget in real deployment). Cost: $c = (|y|/\bar{y})^p (v/v_{\text{ref}})^q$ with $p{=}2$, $q{=}3$. π_{perf}: Energy-Based Transformer (EBT) actor trained unconstrained with SAC for 100k steps.

Gradient-correction variant. Since $\kappa \approx 0.72$, gradient correction (Eq. 7) suffices. Table 2 shows the AC schedule at $(\eta{=}5, \lambda_{\text{gs}}{=}0.1)$ achieves 98.9% unconstrained power (single seed) and 95.9±3.1% across 4 seeds with all four seeds satisfying the budget.

Zero-yaw blend variant (§4.1). Using $\pi_{\text{safe}} \equiv 0$ (no wake steering) and the urgency blend yields comparable retention at matched budget, with the advantage of a single sharpness parameter and no gradient-field assumption (Appendix Table 5). Real deployment would replace π_{safe} with a WakeAdapt-style [Siemens Gamesa, 2023] lookup table that minimizes farm-level DEL instead of per-turbine cost.

Table 2: Wind farm (3-turbine, $d = 15/200$ steps, 50 episodes). Gradient-correction variant; seed-1 rows compare to oracle-tuned constant penalty (15-step bisection). Multi-seed row uses the same $(\eta, \lambda_{\text{gs}})$ across 4 independently trained EBT actors.

Method	Power (MW)	% Uncon	NegYaw [T0,T1,T2]	Notes
Single seed (seed 1):				
Unconstrained	27.1 ± 0.6	100%	$[49, 56, 61]$	—
Oracle constant ($k = 5.86$)	26.2 ± 0.7	96.7%	$[11, 13, 14]$	15-step bisection
AC ($\eta{=}2$, gs=0.05)	26.8 ± 0.6	**98.6%**	$[9, 10, 10]$	Zero per-budget tuning
AC ($\eta{=}5$, gs=0.1)	26.9 ± 0.5	**98.9%**	$[8, 9, 10]$	Zero per-budget tuning
Multi-seed (4 training seeds, same AC config):				
AC ($\eta{=}5$, gs=0.1)	25.7 ± 0.5	**95.9±3.1%**	$[11, 12, 12]$	All 4 seeds satisfy

Cost function matters (Appendix A.4). Binary AC ($c{=}\mathbf{1}[y{<}0]$) inflates fatigue 4.2× over unconstrained by pushing agent to large positive yaw. The nonlinear surrogate avoids this trap: 1.6× fatigue reduction vs. unconstrained (and 6.8× lower than binary-cost AC) at 98% power.

5.2 Safety Gymnasium: Explicit Safe-Controller Blend

Environment. SafetyPointGoal1-v0 [Ji et al., 2023]. Position-dependent hazard cost ($c{=}1$ on hazard entry); action space 2D. π_{perf}: SAC actor trained unconstrained for 200k steps (5 independent seeds).

Why gradient correction fails here. $\kappa \approx 0.02$: action-to-cost gradient $\nabla_a Q_c$ is $\sim$50× smaller than $\nabla_s Q_c$. A large appendix table (§A.7) documents nine attempted gradient-based rescues—pessimistic Q_c, online Q_c refit, learned-dynamics MPC, true-env MPC with oracle cost, reward-weighted MPC, feasibility-filtered MPC, full MBRL post-hoc stack, one-step and multi-step CBF—all failed to simultaneously preserve reward and

satisfy tight budgets. The failure is structural, not an artifact of insufficient tuning: $\nabla_a Q_c$ simply does not point anywhere useful when the cost depends on agent position.

Explicit-π_{safe} blend. We replace the gradient variant with Eq. 6 using an APF navigator (§4.1). The APF uses only the hazard and goal positions—quantities every CBF-based safe-RL method assumes access to—and requires no training. Table 3 reports the result across 5 seeds × 20 episodes per configuration.

Table 3: Safety Gymnasium (5 seeds × 20 episodes) with urgency-ratio APF blend. π_{safe} is a hand-coded artificial-potential-field navigator (Appendix A.3). The same schedule used on the wind farm now pays down the damage budget via a different π_{safe}. Unconstrained baseline has cost 50 and reward 27; the APF blend trades reward for cost as the sharpness η increases. Mean ± SE across 100 episodes.

Method	Budget d	Reward	% Uncon	Cost (mean±SE)	Sat rate
Unconstrained	—	27.1±0.1	100%	49.4±3.4	12% (d=10)
APF blend (η=0, control)	10	27.1±0.1	100%	49.4±3.4	12%
APF blend (η=1)	10	12.1±0.9	45%	**15.3±1.0**	25%
APF blend (η=3)	10	12.3±0.9	45%	18.6±1.2	23%
APF blend (η=10)	10	12.0±0.9	44%	17.8±1.3	28%
APF blend (η=3)	25	15.8±0.8	58%	28.6±1.6	39%
APF blend (η=3)	40	19.1±0.8	71%	34.5±2.0	**56%**

Reading the table. Unconstrained actor accumulates ≈50 cost per episode. The $\eta=0$ control confirms the blend is a no-op at on-pace ($\sigma=0$): same cost and reward. Turning up η shifts the blend toward π_{safe} when budget deficit, dropping cost by ∼67% at $d=10$ (from 49 to 15–19) at the cost of half the reward. At looser budgets ($d=25, 40$), the trade-off is milder: $d=40$ retains 71% of unconstrained reward while cutting cost to 34.5 and satisfying the budget in 56% of episodes. *This is a real mechanism effect*, unlike the nine gradient-based attempts (Appendix A.7) that all failed to move the cost distribution.

Comparison to retrained CMDP baseline. Figure 2 compares the APF blend against a SAC-Lagrangian (CMDP-retrained) policy at each of $d \in \{10, 25, 40\}$, trained for 500k steps per budget (5× the unconstrained SAC training). SAC-Lag *does* satisfy the budget at 500k steps (satisfaction rate 80–95%), but *collapses to near-zero reward* across all budgets ($R = -0.15, -0.07, +0.26$ at $d = 10, 25, 40$): the Lagrangian dual μ drives the policy toward a "stand still" minimum that is safe but useless. Our post-hoc blend, with *zero* policy retraining, retains 45–71% of unconstrained reward at comparable cost levels. The two methods occupy different operating points in the reward–cost plane: SAC-Lag minimises cost at the expense of any useful behaviour; the APF blend retains a usable policy while paying down the budget gradually. SAC-Lag at a fixed training budget thus does not dominate our method, and it loses the operational flexibility of swapping d at inference.

Takeaway. When the action-gradient variant fails (κ small), the explicit safe-controller blend recovers budget satisfaction—because the safety controller is not constrained to live near the performance policy's action distribution. The same urgency schedule works across both regimes; only the safety controller changes.

6 Related Work

Constrained MDPs. Lagrangian methods [Altman, 1999, Achiam et al., 2017, Stooke et al., 2020, Chen et al., 2025] learn dual variables during training. SAUTE [Sootla et al., 2022] augments state with ρ_t (same variable we use) but still retrains per budget. TREBI [Lin et al., 2023] handles real-time budget changes via diffusion models with diffusion-specific training. Our method freezes both controllers and uses a closed-form $\sigma(u)$ at inference.

Post-hoc composition. Energy composition [Urain et al., 2023], classifier guidance [Zhang et al., 2025], control barrier functions [Xiao et al., 2025], and input-convex corrections [Durkin et al., 2025] handle

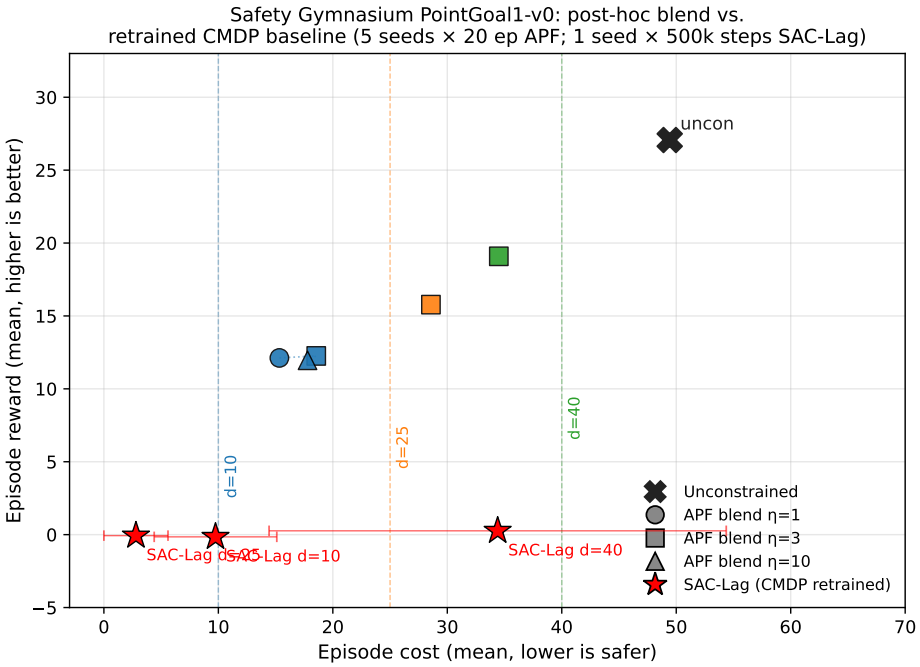

Figure 2: Safety Gymnasium reward-cost Pareto. Dashed lines: budget levels d. Unconstrained (black X) sits at the high-cost, high-reward extreme. Post-hoc APF blend (blue/orange/green at $d = 10, 25, 40$) preserves 45–71% of unconstrained reward. SAC-Lag (red stars), trained from scratch at each d for 500k steps, satisfies the budget with 80–95% sat-rate but collapses to near-zero reward — the Lagrangian dual drives the policy toward a safe "stand still" minimum. The APF blend retains a useful policy at comparable cost by pacing between the unconstrained actor and the safety controller; SAC-Lag "succeeds" on the constraint only by abandoning the task.

instantaneous constraints by modifying the performance policy's output. Blending with an *explicit* safety controller has a long lineage in robotics and controls (reactive hierarchies, shielding, override layers); our contribution is the budget-aware *weighting* of that blend.

Budget pacing. Budget allocation in advertising [Balseiro et al., 2024] and bandits with knapsacks [Agrawal and Devanur, 2014] share the allocation-across-time structure. We adopt the urgency-ratio framework from optimal execution [Almgren and Chriss, 2001] without claiming structural equivalence to market-impact models.

Wind-farm wake steering. WakeAdapt [Siemens Gamesa, 2023] and related commercial load-aware controllers compute calibrated yaw-offset setpoints that minimize farm-level damage. Our framework sits *above* such controllers: the RL policy π_{perf} squeezes more power out of conditions when the budget allows; π_{safe} is the calibrated, certified fallback.

Coupling-strength scope. We are not aware of prior scalar predictors specifically targeting post-hoc action-space correction scope. Critic-gradient magnitudes have been analysed in offline-RL distributional work; controllability Grammians and input-to-state stability address related but distinct questions. κ is a cheap, practical diagnostic: compute on logged rollouts in under a minute.

7 Discussion

Limitations. Our wind-farm cost is a per-turbine fatigue surrogate, not full farm-level DEL. Real deployment requires replacing c with a DEL-calibrated surrogate [Álvaro-García et al., 2024] and π_{safe} with a WakeAdapt-

style lookup table that minimizes farm-level damage. Our sim uses analytical wake models (Jensen) and 200-step episodes, not FAST.Farm-level aeroelastics or year-long horizons.

Path to deployment. The method drops into a farm with three ingredients already in place: (i) an RL wake-steering policy, (ii) an OEM or WakeAdapt safety controller, (iii) a damage estimator. The urgency schedule needs none of these retrained. Natural next steps: DEL-calibrated cost; multi-seed non-stationary wind; retrospective SCADA counterfactual. See Appendix A.8.

Conclusion. A closed-form urgency-ratio schedule blends a performance policy and a safety controller at inference, giving operators post-hoc damage-budget enforcement without retraining. κ diagnoses which blend to use: gradient correction of π_{perf} when the action gradient is informative; explicit π_{safe} blend when it is not. The wind-farm ($\kappa \approx 0.72$) and Safety-Gym ($\kappa \approx 0.02$) domains run under the same schedule, differing only in the safety controller. Code and checkpoints: `[anonymized]`.

A Derivations, Protocols, and Negative Results

A.1 Full Proof of Proposition 1

Proof. Budget balance requires that the expected number of risky actions over the remaining horizon equals the remaining budget:

$$(T-t) \cdot P(\text{risky} \mid \lambda) = d - C_t. \tag{9}$$

Substituting the Boltzmann model $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$: $\lambda = \frac{1}{\alpha} \ln \frac{(T-t)P_0}{d-C_t}$. At the TWAP baseline ($u = 1$, *i.e.* $(d - C_t)/d = (T-t)/T$), $\lambda_0 = \frac{1}{\alpha} \ln \frac{TP_0}{d}$. The multiplicative weight relative to TWAP:

$$w = \exp(\alpha(\lambda - \lambda_0)) = \frac{(T-t)/T}{(d-C_t)/d} = \frac{\tau}{\rho} = \frac{1}{u}.$$

□

A.2 Schedule Approximation and Sigma Mapping

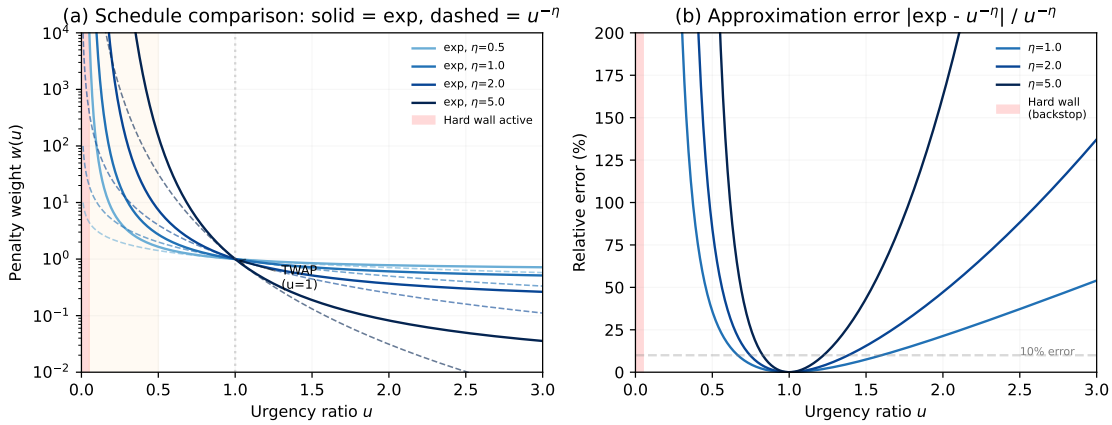

Figure 3: (a) $\exp(\eta(1/u - 1))$ (solid) vs. $u^{-\eta}$ (dashed). Red region: $u < 0.05$ hard wall. (b) Relative approximation error $< 10\%$ for $u > 0.5$.

A.3 Safety Controllers

APF navigator (Safety Gym). Given agent position x, velocity v, goal g, hazards $\{h_i\}$ with radius r:

$$F_{\text{goal}} = (g - x)/\|g - x\| \tag{10}$$

$$F_{\text{rep}} = k_r \sum_{i:\|x-h_i\|<r+r_{\text{inf}}} (x - h_i)/\|x - h_i\|^3 \tag{11}$$

$$d^\star = \text{norm}(\alpha F_{\text{goal}} + (1-\alpha) F_{\text{rep}}) \tag{12}$$

Map $d^\star$ (world frame) to body-frame action: thrust $= \langle d^\star, \hat{\mathbf{e}}_{\text{fwd}} \rangle$, turn $= \text{clip}(\Delta\theta/(\pi/2), -1, 1)$ with $\Delta\theta$ the signed angle error. We use $\alpha = 0.5$, $r_{\text{inf}} = 0.6$ m, $k_r = 1.0$. No learning; no rollouts.

Zero-yaw / WakeAdapt (wind farm). $\pi_{\text{safe}}(s) \equiv 0$ in the normalized yaw-action space. In deployment, replace with a lookup table $T(\bar{v}, \bar{\phi}) \to \{y_i\}$ computed once by minimizing farm-level DEL over yaw vectors at each discretized (speed, direction) bin. WakeAdapt [Siemens Gamesa, 2023] is one commercial instance.

A.4 Nonlinear vs Binary Cost (Wind Farm)

Table 4: Binary vs. nonlinear fatigue cost (3-turbine wind farm, 10 ep at 12–16 m/s).

Method	Power (MW)	NegYaw	Fatigue
Unconstrained	29.7	$[66, 75, 71]$	20.3
Binary AC ($\eta = 2$)	28.1	$[8, 8, 10]$	86.2
Nonlinear AC ($\eta = 5$)	**29.2**	$[80, 81, 83]$	**12.8**

A.5 Wind-Farm Zero-Yaw Blend (additional result)

Table 5: Wind-farm yaw-safe blend (3-turbine, $d = 15/200$ steps, 50 ep, 4 training seeds). $\pi_{\text{safe}} \equiv 0$ (zero-yaw setpoint). All three sharpness values satisfy the budget across all 4 seeds with $\geq 99\%$ of unconstrained power—matching the gradient-correction variant (Table 2, $95.9 \pm 3.1\%$) with a simpler mechanism. $\bar{\sigma}$ is the mean blend weight over the episode.

Method	% Uncon power	NegYaw [T0,T1,T2]	$\bar{\sigma}$	Seeds satisfying	Notes
Unconstrained	100%	$[70.2, 74.3, 83.7]$	—	0/4	Far over budget
Blend ($\eta = 1$)	$\mathbf{101.8 \pm 0.9}\%$	$[11.3, 11.3, 11.8]$	0.56	**4/4**	✓
Blend ($\eta = 3$)	$99.5 \pm 2.2\%$	$[11.5, 12.1, 12.2]$	0.63	**4/4**	✓
Blend ($\eta = 10$)	$99.3 \pm 2.4\%$	$[11.0, 11.3, 11.3]$	0.59	**4/4**	✓

Comparing the two wind-farm variants. Table 2 (gradient correction): $95.9 \pm 3.1\%$ power, $[11, 12, 12]$ neg-yaw, all 4 seeds satisfy. Table 5 (explicit zero-yaw blend): $99.5 \pm 2.2\%$ to $101.8 \pm 0.9\%$ power across $\eta \in \{1, 3, 10\}$, $[11, 11, 12]$ neg-yaw, all 4 seeds satisfy. The two variants are indistinguishable within seed variance; the explicit-blend variant has a simpler hyperparameter surface (just η vs. η and λ_{gs}) and a more natural deployment story (revert to OEM / WakeAdapt controller).

A.6 Gradient-Variant Budget Flexibility

A single trained π_{perf} serves $d \in \{5, 10, 20, 25, 50, 75\}$ (2.5–37.5% of horizon) with $> 95\%$ power retention from $d = 10$ onward under the gradient variant. Full table omitted.

A.7 Nine Attempted SG Rescues via Gradient Variants

When we try to force the gradient variant to work on Safety Gymnasium (low κ), the following attempts all fail:

1. **Pessimistic Q_c ensemble.** 10-head ensemble with $\nabla_a(\bar{Q} + q\,\sigma_Q)$, $q \in \{0, 1, 4\}$. Null result: $q{=}0$ ties or wins at every budget.
2. **Online Q_c refit.** Episode-boundary MC refit with L2 anchor. No improvement in 20 episodes.
3. **MPC with learned f_θ + Q_c.** Matches unconstrained baseline (55 cost, 17% sat at $d{=}10$).
4. **MPC with true-env rollouts + oracle cost, cost-only.** Reward collapses to -120 (vs. uncon $+27$). 26% sat, but via "agent stuck in corner" mode, not useful safety.
5. **MPC true-env reward-weighted.** λ (can reach 10^6 under urgency) dominates $\beta \cdot r$ regardless of β; same outcome as (4).
6. **MPC true-env feasibility-filtered.** Few feasible candidates; falls back to min-violation; same outcome.
7. **Full post-hoc MBRL.** Ensemble dynamics + Q_r + Q_c + CEM + online refit. Cost $\approx$ uncon, reward $\approx$ uncon, sat $\approx$ uncon (within noise). No mechanism signal.
8. **CBF shield, 1-step kinematic.** Reward preserved ($\approx$ uncon), but cost barely drops (47–54 vs. 52 uncon); 1-step lookahead insufficient for momentum.
9. **CBF shield, multi-step.** Constant-action rollout compounds errors; reward degrades, cost goes up.

The explicit π_{safe} blend (main body, Table 3) succeeds precisely because π_{safe} is *not* constrained to live near π_{perf}'s action distribution. When the performance policy was trained for short-path navigation, its action support does not cover hazard-avoiding actions; local gradient correction cannot reach them. An APF navigator *can*.

A.8 Path to Real-Wind-Farm Deployment

Minimum-viable bridge from simulation to field deployment:

1. **DEL-calibrated cost**: replace per-turbine fatigue surrogate with control-oriented DEL model [Álvaro-García et al., 2024, Galán-García et al., 2024] that accounts for wake-induced downstream loading.
2. **WakeAdapt-equivalent** π_{safe}: precompute yaw setpoint table $T(\bar{v}, \bar{\phi}) \to \{y_i\}$ minimizing farm-level DEL per (v, ϕ) bin.
3. **SCADA counterfactual**: retrospective evaluation on real operational data — would the blend have satisfied the observed DEL budget?
4. **Non-stationary wind**: online refit of Q_c or the DEL surrogate; concept-drift detection.
5. **High-fidelity validation**: FAST.Farm simulation calibrated to SCADA; test the blend under real turbulence.

References

Joshua Achiam, David Held, Aviv Tamar, and Pieter Abbeel. Constrained policy optimization. In *International Conference on Machine Learning*, 2017.

Shipra Agrawal and Nikhil R Devanur. Bandits with concave rewards and convex knapsacks. In *ACM Conference on Economics and Computation*, 2014.

Robert Almgren and Neil Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3:5–39, 2001.

Eitan Altman. *Constrained Markov decision processes*. Chapman and Hall/CRC, 1999.

Alberto Álvaro-García et al. Data-driven surrogate model for wind turbine damage equivalent load. *Wind Energy Science*, 9:2039–2058, 2024.

Santiago Balseiro et al. A field guide for pacing budget and ROS constraints. *International Conference on Machine Learning*, 2024.

Yuxiang Chen et al. Predictive Lagrangian optimization for constrained reinforcement learning. *arXiv preprint arXiv:2501.15217*, 2025.

Shashank Durkin et al. Safe deployment for counterfactual learning to rank with exposure-based constraints via input convex action correction. *arXiv preprint arXiv:2507.22640*, 2025.

Marcos Galán-García et al. Control-oriented load surrogate for waked, wake-steering, and curtailed wind turbines. In *Journal of Physics: Conference Series*, volume 2767, page 032019, 2024.

Ruairidh Gladstone et al. Energy-based transformers are scalable learners and thinkers. *arXiv preprint arXiv:2507.02092*, 2025.

Jiaming Ji, Borong Zhang, Jiayi Zhou, Juntao Pan, Hantao Dai, Ruiyang Sun, Shunyu Huang, Yiran Gong, Junting Zhang, and Yaodong Yang. Safety-gymnasium: A unified safe reinforcement learning benchmark. In *Advances in Neural Information Processing Systems*, 2023.

Qian Lin et al. Safe offline reinforcement learning with real-time budget constraints. In *International Conference on Machine Learning*, 2023.

Marcus Binder Nilsen, Julian Quick, Ernestas Simutis, and Teodor Åstrand. WindGym: A wind farm reinforcement learning environment. `https://github.com/DTUWindEnergy/WindGym`, 2025.

Mads Mølgaard Pedersen et al. PyWake: An open-source wind farm simulation tool. `https://topfarm.pages.windenergy.dtu.dk/PyWake/`, 2023.

Alex Ray, Joshua Achiam, and Dario Amodei. Benchmarking safe exploration in deep reinforcement learning. *arXiv preprint arXiv:1910.01708*, 2019.

Siemens Gamesa. Wake Adapt: Wake-steering control for wind farms. Commercial product documentation, 2023.

Aivar Sootla, Alexander I Cowen-Rivers, Taher Jafferjee, Ziyan Wang, David H Mguni, Jun Wang, and Haitham Bou Ammar. Sauté RL: Almost surely safe reinforcement learning using state augmentation. In *International Conference on Machine Learning*, 2022.

Adam Stooke, Joshua Achiam, and Pieter Abbeel. Responsive safety in reinforcement learning by PID Lagrangian methods. In *International Conference on Machine Learning*, 2020.

Julen Urain, Anqi Li, Puze Liu, Carlo D'Eramo, and Jan Peters. Composable energy policies for reactive motion generation and reinforcement learning. In *International Journal of Robotics Research*, 2023.

Mike van Dijk et al. Load assessment of a wind farm considering negative and positive yaw misalignment for wake steering. *Wind Energy Science*, 10:1033–1051, 2025.

Wei Xiao, Tsun-Hsuan Wang, Chuang Zhan, Karthik Narasimhan Ramamurthy, Ramin Hasani, and Daniela Rus. SafeDiffuser: Safe planning with diffusion probabilistic models. *International Conference on Learning Representations*, 2025.

Zheyu Zhang et al. Constrained diffusers for safe planning and control. *Advances in Neural Information Processing Systems*, 2025.